Rev A wiring schematic / notes

MIDI DIN IN -> 6N138 optocoupler -> ESP32-C3 SuperMini GPIO20 (UART RX, 31250 baud)
ESP32-C3 SuperMini GPIO21 (UART TX, 115200 baud) -> Cmod A7 uart_rx

This sheet is a KiCad starter schematic. Before PCB production, replace the notes with exact symbols/footprints for the DIN connector and your exact ESP32-C3 SuperMini clon

MIDI input side

J1 DIN-5:
Pin 4 -> R1 220R -> 6N138 LED anode
Pin 5 -> 6N138 LED cathode return
Pin 2 -> cable shield/chassis only
Pins 1,3 -> NC

Keep the MIDI DIN side isolated from ESP32/Cmod signal ground.

Logic side

U2 6N138 powered from ESP32 3V3
C1 100nF near U2
R2 2.2k..4.7k pull-up from opto output to 3V3
Opto output -> ESP32-C3 GPIO20

Optional LED activity indicator can be driven from ESP32 firmware after the first bring-up.

Cmod A7 header

ESP32 GPIO21 / UART TX -> Cmod uart_rx
ESP32 GND -> Cmod GND
3V3 only if intentionally sharing power
Optional ESP32 GPIO20/other RX <- future Cmod TX

For first power-up, power ESP32-C3 from USB and share only GND + TX signal with Cmod.

Default MIDI note map in firmware

36 kick -> '1'
38 snare -> '2'
42 closed hihat -> '3'
46 open hihat -> '4'
39 clap -> '5'
45 low tom -> '6'
50 high tom -> '7'
49 crash -> '8'

Chromatic fallback: notes 36..43 -> pads 1..8.

ESP32-C3 SuperMini MIDI bridge
CMOD A7 Drum Computer
Sheet: /
File: midi_esp32_c3_supermini_adapter.kicad_sch

Title: MIDI DIN to Cmod A7 Adapter

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